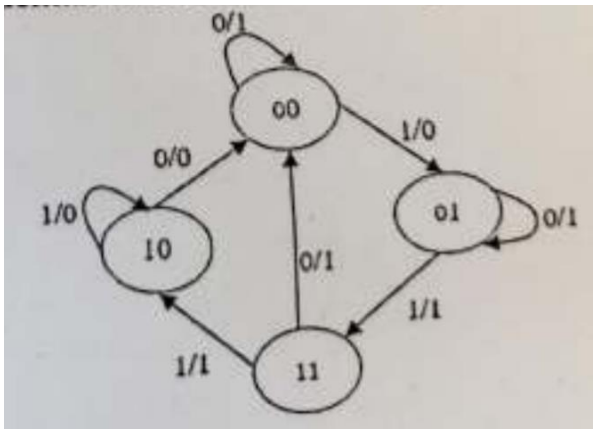
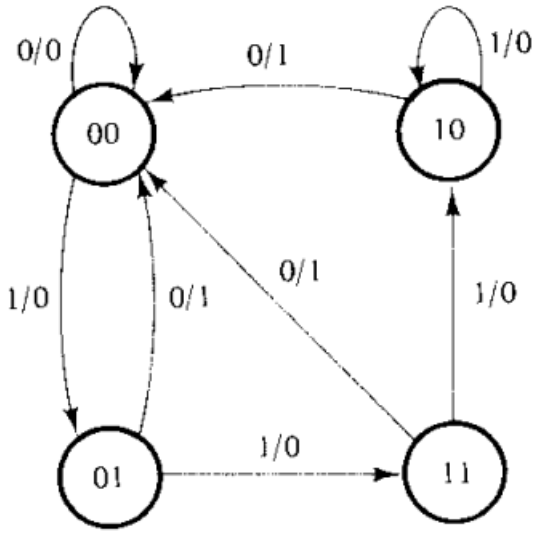
Digital Logic System

Assignment 4

1. What is flip-flop? Explain the operation of SR flipflop with the help of logic circuit and truth table.
2. What is JK flip-flop? Explain its operation with the help of logic circuit and truth table.
3. What is T flip-flop? Explain its operation with the help of logic circuit and truth table.
4. What is D flip-flop? Explain its operation with the help of logic circuit and truth table.
5. Explain about state diagram and state table.
6. Explain the operation of master slave flip-flop with diagram.
7. Design a synchronous up/down counter.
8. Explain the operation of serial in parallel out (SIPO) shift register with the help of necessary circuit diagram.
9. Design a BCD ripple counter.
10. Explain the operation of asynchronous decade counter with timing diagrams.
11. Why decoder and counter are respectively called combinational and sequential circuits?
12. Design a 3-bits binary counter using T flipflop.
13. Explain the operation of parallel in serial out (PISO) shift register with the help of necessary circuit diagram and timing diagram.
14. Differentiate between latch and flip-flop.
15. Design a 4-bit synchronous up counter with JK flip-flops.
16. Convert the following state diagram into state table.